Om Shete
THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE
BE C31
2103163

7

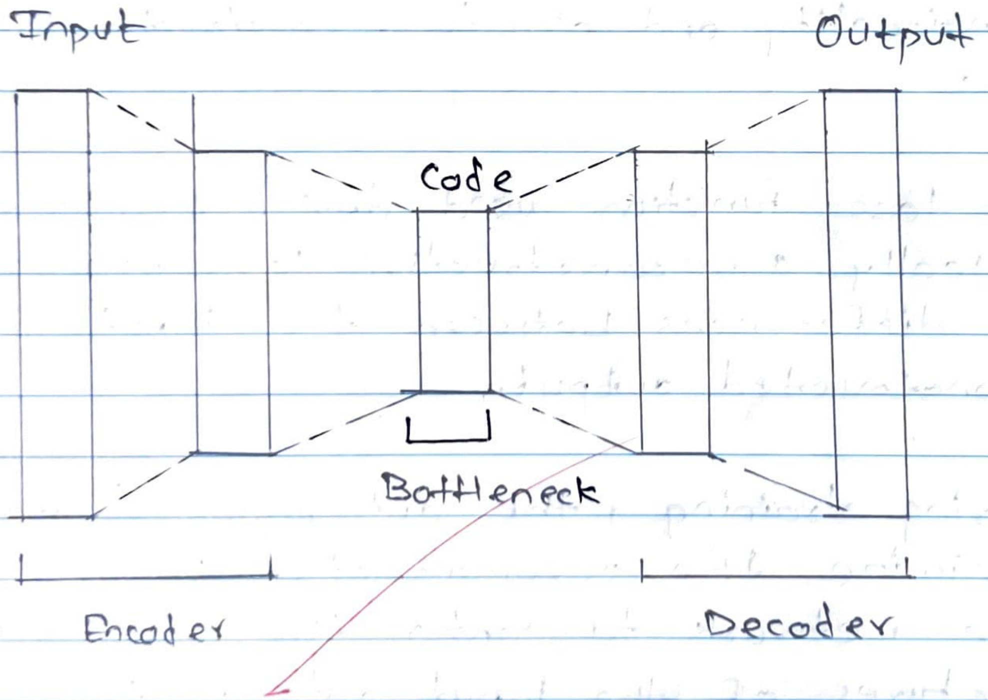
Assignment - 5

Topic : Autoencoder

Q.1 Explain the autoencoder architecture with appropriate block diagram.

=>

- Autoencoders are simple learning networks that aim to transform inputs into outputs with the minimum possible error.
- They follow an encoder-decoder architecture.
- The encoder maps the input to latent space and decoder reconstructs the input.



- The latent space has lower dimension than the input. The low dimensional latent variables can encode most important features of the input and they are capable of reconstructing it.
- The architecture of autoencoder is presented above.

1) Encoder

⇒ Input layer takes raw input data. The hidden layers progressively reduce the dimensionality of the input, capturing the important features and patterns. These layer compose the encoder.

2) Decoder

⇒ The bottleneck layer takes the encoded representation and expands it back to the dimensionality of original input.

- The hidden layers progressively increases the dimensionality and aim to reconstruct the original input.

3) The loss function used during training is typically a reconstructive loss, measuring the differences between the input and reconstructed output.

4) During training, the autoencoder learns to minimize the reconstruction loss, forcing the network to capture the most important features of the input data in the bottleneck layers.

- After training process, only the encoder part of autoencoder is retrained to encode a similar type of data used in training process.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

Q.2 What are sparse autoencoder? Explain sparsity parameter and how does the network enforce the average activation value to be equal to sparsity parameter.

=)

- A sparse autoencoder is simply an autoencoder whose training criterion involves a sparsity constraints or penalty $\Omega(h)$ on the code layer h, in addition to the reconstruction error :

$$L(y, g(f(y))) + \Omega(h)$$

- Sparse autoencoder are typically used to learn features for another task such as classification.
- An autoencoder that has been regularized to be sparse must respond to unique statistical features of the dataset it has been trained on, rather than simply acting as an identity function.
- In this way, training to perform the copying task with a sparsity penalty can yield a model that has learned useful features as a byproduct.
- The penalty $\Omega(h)$ simply as a regularizer term added to a feedforward network whose primary task is to copy the input and possibly also perform some supervised task that depend on these sparse features.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

- How does the Network Enforce Sparsity?

1) KL divergence

⇒ The backpropagation process adjusts the network weights to reduce the KL divergence loss, ensuring neurons are selectively activated.

2) L_1 Regularization

⇒ Sometimes, L_1 regularization is used to push most neurons activation to zero, enforcing sparsity.

3) ReLU

⇒ These functions help control neuron activations within the desired range.

Q.3 Write short note on contractive autoencoders.

⇒

- Another strategy for regularizing an autoencoder is to use a penalty Ω as in sparse auto-encoder,

$$L(y, g(f(y))) + \Omega(h, y).$$

but with a different form of Ω

- An autoencoder regularized in this way is called contractive autoencoder.
- This approach has theoretical connections to denoising autoencoders, manifold learning and probabilistic modeling.

THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

- The contractive autoencoder introduces an explicit regularizer on the code $h = f(y)$, encouraging the derivatives of 'f' to be as small as possible.
- The penalty $\Omega(h)$ is the squared Frobenius norm of the Jacobian matrix of partial derivatives associated with the encoder function.
- There is a connection between the denoising autoencoder and a contractive autoencoder.
- When using the Jacobian-based contractive penalty to pretrain features $f(y)$ for use with a classifier, the best classification accuracy usually results from applying the contractive penalty $f(y)$ rather than to $g(f(y))$.
- A contractive penalty on $f(y)$ also has close connections to score matching.
- The name contractive arises from the way that the CAE warps space.
- One practical issue with the CAE regularization criterion is that although it is cheap to compute in the case of single hidden layer autoencoder, it becomes much more expensive in the case of deeper autoencoder.
- The stratergy followed can be to separately train a series of single-layer autoencoders, each trained to reconstruct the previous autoencoder's hidden layer.

THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

Q.4 Explain regularization in Autoencoders.
=)

- Regularization in autoencoder helps improve their ability to learn meaningful representations while preventing overfitting.
- Regularization techniques modify the training process to ensure that the learned latent representations are robust and generalizable.
- A regularized autoencoder can be non-linear and overcomplete but still learn something useful about the data distribution even if the model capacity is greater enough to learn a trivial identity function.
- In addition to the methods described here which are most naturally interpreted as regularized autoencoders, nearly any generative model with latent variables and equipped with an inference procedure may be viewed as a particular form of autoencoder.
- Two generative modeling approaches that emphasize this connection with autoencoders are the descendants of the 'Helmholtz machine' such as the variational autoencoder and the generative stochastic networks.
- These models naturally learn high-capacity, overcomplete encodings of the input and do not require regularization for these encodings to be useful.
- Their encodings are naturally useful because the models were trained to approximately

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

maximize the probability of the training data rather than to copy the input to the output.

Q.5 Explain Undercomplete autoencoder.

=>

- An autoencoder whose code dimension is less than the input dimension is called as undercomplete.
- It aims to learn from compressed representations.
- The learning process is described simply as minimizing a loss function

$$L(y, g(f(y))).$$

where,

L = loss function penalizing g(f(y)) for being dissimilar from y.

f = nonlinear encoder function

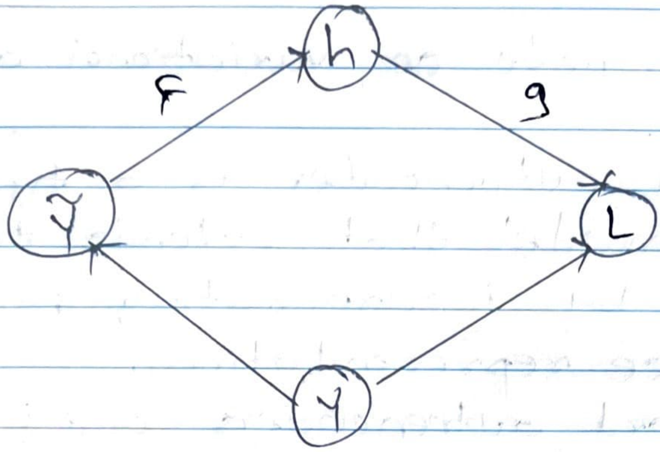
g = nonlinear decoder function

- When the decoder is linear and L is the mean squared error, an undercomplete autoencoder learns to span the same subspace as PCA.
- In this case, an autoencoder trained to perform the copying task has learned the principal subspace of the training data as a side-effect.

- Autoencoders with non-linear encoder functions f and nonlinear ~~trained to perform~~ decoder function g can thus learn more powerful non-linear generalization of PCA.
- Unfortunately, if the encoder and decoder are allowed too much capacity, the autoencoder can learn to perform the copying task without extracting useful information about the distribution of the data.
- Learning an undercomplete representation forces the autoencoder to capture the most salient features of the training data and it won't be able to directly copy its inputs to the output.

Q.6 Explain Denoising autoencoder.
=>

- The denoising autoencoder is an autoencoder that receives a corrupted data as input and is trained to predict the original, uncorrupted data as its output.
- This way the autoencoder can't simply copy the input to its output because the input also contains random noise.
- The noise is substracted and the underlying meaningful data is produced.
- This is called a denoising autoencoder.
- The DAE training procedure is illustrated.



- The computational graph of the cost function for a denoising autoencoder, which is trained to reconstruct the clean data point y from its corrupted version $\tilde{Y}$.
- This is accomplished by minimizing the loss $L = -\log p_{decoder}(y \mid h = f(\tilde{y}))$,
- Typically, the ~~corrupted com~~ distribution $p_{decoder}$ is a factorial distribution whose mean parameters are emmitted by a feed-forward network g.
- The autoencoder then learns a reconstruction distribution $p_{reconstruct}(\tilde{Y} \mid Y)$ estimated from training pairs $C(\tilde{Y}, Y)$ as follows:

1) Sample training example y from the training data.
2) Sample a corrupted version $\tilde{y}$ from $C(\tilde{Y} \mid Y = y)$
3) Use $(Y, \tilde{Y})$ as a training example for estimating the autoencoder.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

Q.7 Write short note on variational autoencoder.

⇒

- A variational autoencoder is a type of generative model that extends traditional autoencoders by incorporating probabilistic latent space representations.
- Unlike standard autoencoders, which learn deterministic encodings, VAEs learn a distribution over latent variables, making them useful for generative new data samples.
- Key concepts :-

1) Latent space representation

⇒ Instead of encoding an input into a fixed latent vector, VAEs map inputs to a probability distribution.

- The encoder predicts mean (μ) and variance (σ^2), from which laten variables are sampled.

2) Reparameterization Trick

⇒ Sampling directly from a distribution is non-differentiable, so VAEs use the reparameterization trick :

$$z = \mu + \sigma \cdot \varepsilon \quad , \quad \varepsilon \sim N(0, 1)$$

- This allows gradients to flow during training making backpropagation possible.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

3) Loss Function

⇒ The VAE loss consists of two term:

a) Reconstruction loss — Measures how well the decoder reconstructs the input.

b) KL divergence loss — Ensures the learned latent distribution is close to a standard normal distribution.

$$L = L_{reconstruction} + \beta \cdot D_{KL}(q(z|x) \| p(z))$$

AP
9/4/25